Associations of BMI, Age, Blood Pressure, and Physical Activity With Diabetes Among Adults: An NHANES Analysis

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Code and notebook: https://github.com/joshnimal08-cell/diabetes-analysis

Introduction

Diabetes is common in the United States. The CDC estimates that about 40.1 million people, roughly 12% of the population, had diagnosed or undiagnosed diabetes in 2023, and about 27.6% of adults with diabetes (around 11.0 million people) have not been diagnosed [1].

Public health sources name several risk factors. A higher BMI, physical inactivity, and high blood pressure are linked to a higher risk of type 2 diabetes, and risk rises with age, starting around 45 and increasing sharply after 65 [2]. These factors are usually described one at a time, but they may overlap, so it is useful to look at them together in the same group of people.

In this project, I used data from the NHANES 2017 to March 2020 cycle to examine how BMI, age, blood pressure, and physical activity are associated with diabetes among adults [3].

Methods

Data source. The data come from the National Health and Nutrition Examination Survey (NHANES), which is run by the CDC's National Center for Health Statistics. I used the NHANES 2017 to March 2020 pre-pandemic file, which combines the 2017 to 2018 cycle with data collected from 2019 to March 2020 before COVID-19 stopped fieldwork. NHANES data are publicly available and de-identified, so no additional ethics approval was required for this analysis [3].

Sample. I merged four files (Demographics, Diabetes Questionnaire, Body Measures, and Blood Pressure) using each participant's ID number, then added the Physical Activity Questionnaire. I excluded participants under 18, those whose diabetes answer was

Borderline, Refused, or Don't know, those missing BMI or blood pressure, and those whose activity answer was Don't know. Of the 15,560 participants in the demographics file, 7,720 adults remained in the final analysis.

Variables. Diabetes was defined as a doctor having told the participant they have diabetes. Physical activity was defined as moderate recreational activity for at least 10 minutes in a typical week. Blood pressure was measured using the first systolic reading. Diastolic blood pressure was left out of the tests and the regression because its group averages were nearly identical (74.4 vs. 74.9 mmHg).

Analysis. I used t-tests to compare average BMI, blood pressure, and age between groups, and a chi-square test to compare diabetes rates by physical activity. Finally, I used logistic regression to examine all four variables together. Analyses were done in Python (pandas, scipy, and statsmodels), and results were considered significant at $p < .05$.

Results

The final analytic sample included 7,720 adults from the NHANES 2017 to March 2020 cycle. Of these, 1,167 (15.1%) had diabetes, based on self-reported physician diagnosis.

Adults with diabetes had a significantly higher mean BMI ($M = 32.8$) than adults without diabetes ($M = 29.3$), $t(7718) = 14.87$, $p < .001$ (Figure 1). Adults with diabetes also had significantly higher mean systolic blood pressure ($M = 131.3$ vs. $M = 122.9$), $t(7718) = 13.51$, $p < .001$, and significantly higher mean age ($M = 62.2$ vs. $M = 46.7$), $t(7718) = 27.92$, $p < .001$. Diabetes rates were also significantly higher among adults who reported no regular physical activity (17.6%) than among those who were active (11.7%), $\chi^2(1) = 49.30$, $p < .001$ (Figure 2).

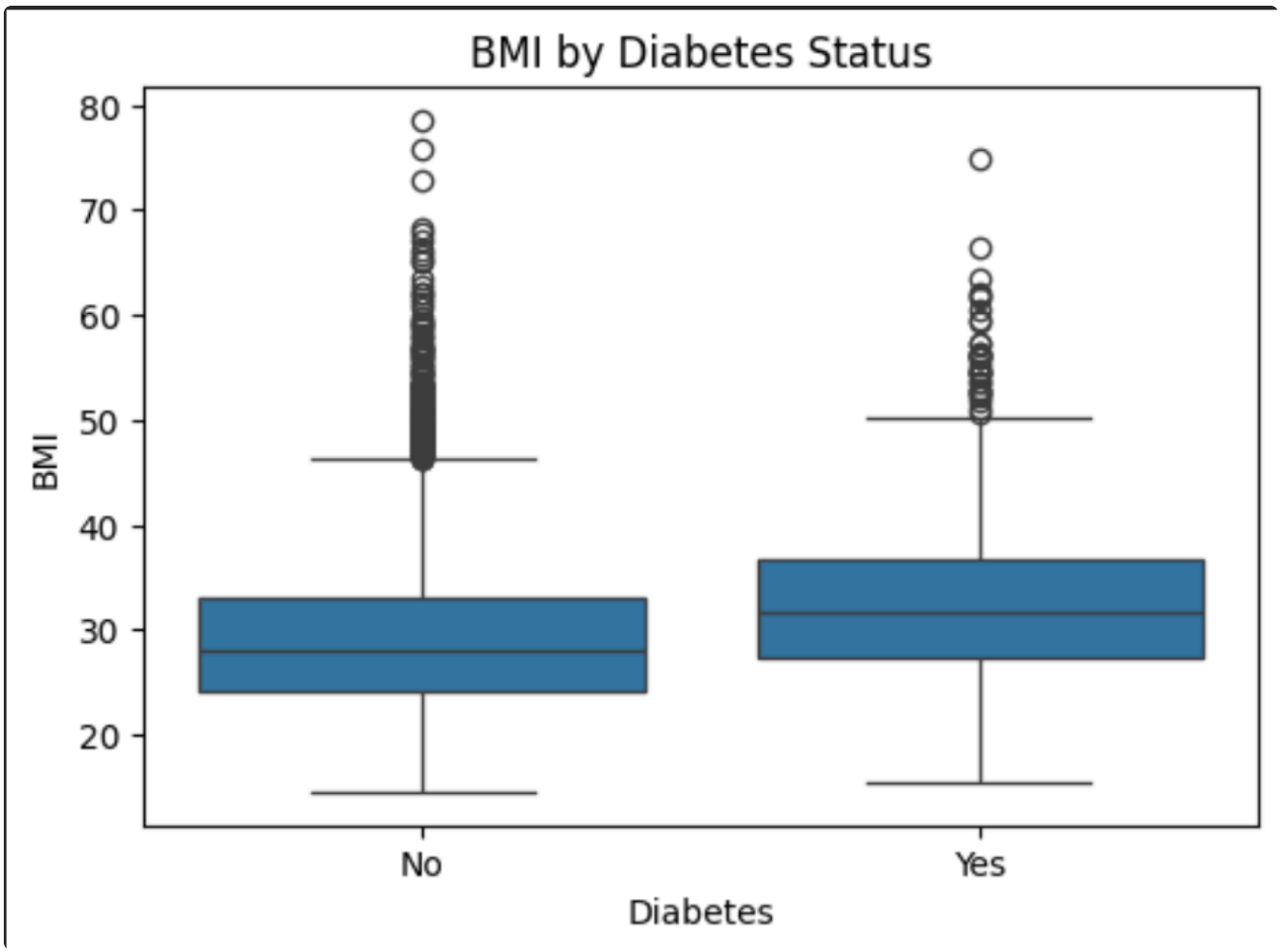


Figure 1. BMI by diabetes status.

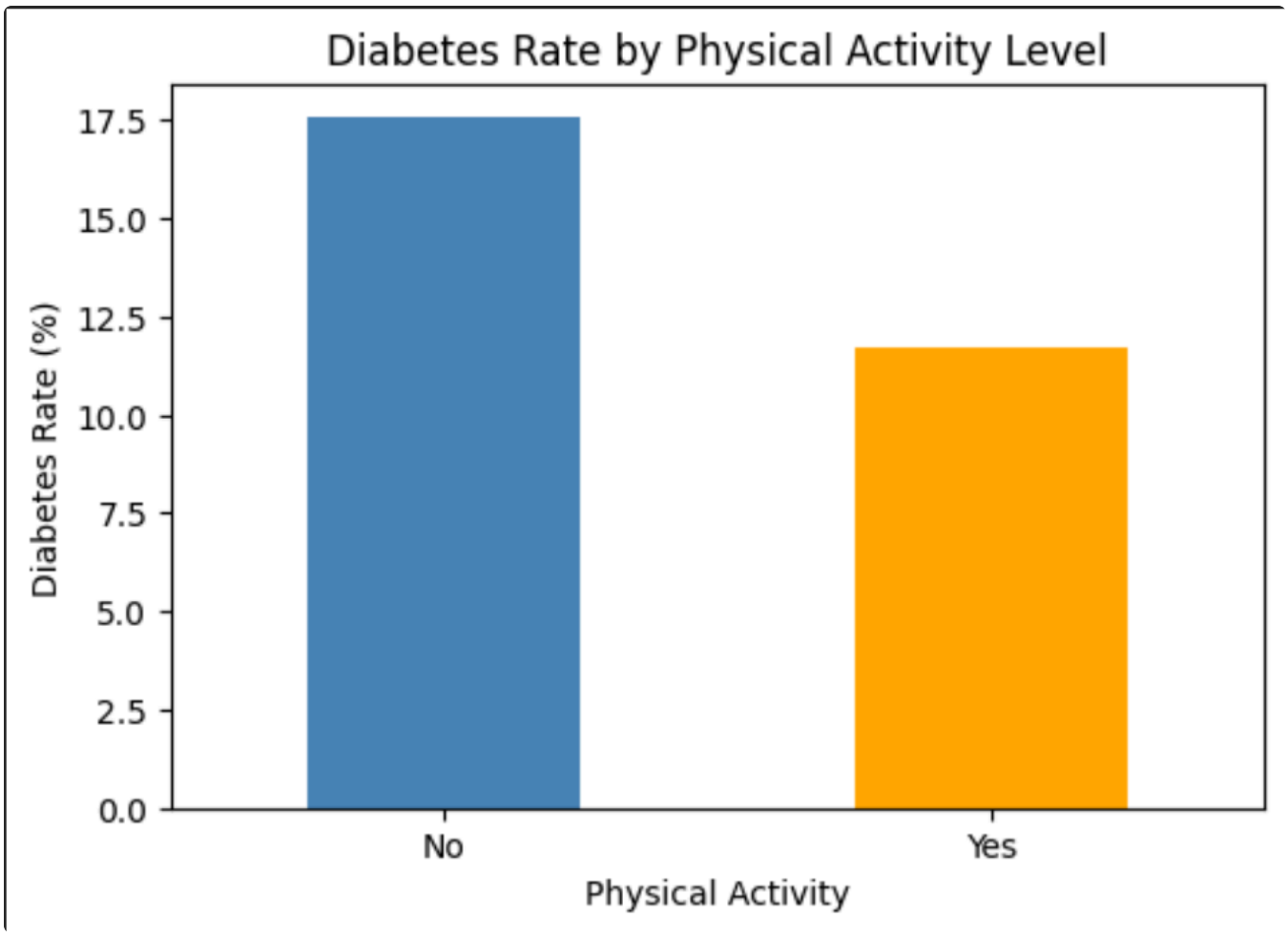


Figure 2. Diabetes rate by physical activity level.

To look at all four variables at the same time, I ran a logistic regression predicting diabetes status using age, BMI, systolic blood pressure, and physical activity together (Table 1). Age was a significant predictor (OR = 1.06, $p < .001$), meaning each additional year of age was linked to about a 6% increase in the odds of having diabetes, even after accounting for the other variables. BMI was also significant (OR = 1.07, $p < .001$), with each one unit increase in BMI linked to about a 7% increase in the odds of diabetes. Physical activity was significant too (OR = 0.79, $p = .001$), but in the opposite direction: adults who reported being physically active had about 21% lower odds of diabetes than those who did not. Interestingly, blood pressure did not remain a significant predictor in this model (OR = 1.002, $p = .271$), even though it had looked significant when I tested it on its own. This suggests that blood pressure's apparent link to diabetes may be explained mostly by its relationship with age rather than an independent effect of its own.

Predictor	Odds ratio	*p*
Age (per year)	1.059	< .001
BMI (per unit)	1.071	< .001
Systolic blood pressure (per mmHg)	1.002	.271
Physical activity (active vs. not active)	0.793	.001

Table 1. Logistic regression predicting diabetes (N = 7,720).

Discussion

This project analyzed how BMI, physical activity, blood pressure, and age are associated with diabetes among adults in the NHANES sample. Each variable was first tested individually for its association with diabetes. When examined together in a logistic regression, blood pressure was not a significant independent predictor of the disease, while age, BMI, and physical activity were.

Interestingly, blood pressure was significantly associated with diabetes when tested on its own, but this association disappeared once age and BMI were included in the same model. In this sample, systolic blood pressure was moderately correlated with age (r = .45) but almost uncorrelated with BMI (r = .04), so the change is most likely due to age. Older adults tend to have both higher blood pressure and a higher chance of diabetes, so some of blood pressure's apparent effect was really age showing up through it. This is a real world example of confounding, where a third variable is related to both the predictor and outcome, making the predictor appear influential when its effect may actually be explained by something else.

The results of this project are consistent with common knowledge about diabetes risk factors. Higher BMI, older age, and lower physical activity were all associated with a higher likelihood of diabetes in this sample. The one less expected finding was that blood pressure's association with diabetes did not hold up once age and BMI were accounted for, suggesting its apparent link to diabetes may be driven more by its relationship with age than an effect of its own.

The boxplot (Figure 1) also showed that although the median BMI is higher for adults with diabetes, a handful of individual outliers with very high BMI appear in the non-diabetes group as well. This does not contradict the overall association, since a statistical

association reflects a pattern on average across the whole group, not a rule that applies to every individual case.

Limitations

Because this study used observational data from a single point in time, it is only able to identify associations between variables, not causal relationships. Diabetes status was based on self-report rather than a blood test, so adults who had diabetes but had never been diagnosed would have been counted as not having it. Because of this, some participants may have been placed in the wrong group, which could make the true differences between groups look smaller than they really are. Physical activity was measured with a single Yes/No question, so it could not tell apart people who were very active from people who were only barely active. It was also self-reported, so some participants may have over or under reported how active they really are.

Other factors that were not included in the analysis, such as diet and family history, may also be related to diabetes and could be influencing the associations we found. Participants with missing data were excluded, and if they differ from those who completed everything, for example if they were sicker, the results may not represent all adults. Finally, NHANES samples some groups more heavily on purpose, and survey weights were not applied in this analysis, so the results describe this sample rather than all U.S. adults.

Conclusion

In this sample of adults, older age and higher BMI were associated with higher odds of diabetes, while physical activity was associated with lower odds. Blood pressure was associated with diabetes on its own, but this association was no longer significant after accounting for age and BMI. Because this study was observational, these findings show associations, not causes. Future research could measure diabetes more accurately by using blood tests instead of self-report.

References

[1] Centers for Disease Control and Prevention. (2026, September 16). *National Diabetes Statistics Report*. https://www.cdc.gov/diabetes/data/statistics-report/

[2] Centers for Disease Control and Prevention. (n.d.). *How your test is scored*. https://www.cdc.gov/diabetes/widgets/risktest/how-your-test-is-scored.html

[3] National Center for Health Statistics. (2021, May). *NHANES 2017 to March 2020 data documentation: Demographic variables and sample weights*. https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/2017/DataFiles/P_DEMO.htm

Additional NHANES component pages used, at the same web address path: P_DIQ.htm (diabetes questionnaire), P_BMX.htm (body measures), P_BPXO.htm (blood pressure), and P_PAQ.htm (physical activity).